



**SIMATS**  
**ENGINEERING**



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

## Skill Progress Analytics with Personalized Learning Path Recommendation Engine

**Course Name :Java Programming**

**Course Code :CSA0925**

Guided by

Dr. Sivasankar C

By

S. Dharanish(192472152)

P. Jeevan Kumar(192472082)

Akileshwaran A(192511354)

# Review-1 Feedback

## Feedback Received

- Need clearer system architecture and module flow.
- Need detailed explanation of database and UML design.
- Need to show implementation progress.
- Need to include UI and testing details.

## Corrective Actions

- Updated the system architecture.
- Defined all major project modules.
- Added UML and ER diagram designs.
- Started implementation of Java and MySQL modules.
- Added UI, testing and reporting components.



# Updated Architecture



## Changes Made

- Added separate modules for student registration and login.
- Added course management and progress tracking.
- Added skill analytics module.
- Added personalized recommendation engine.
- Added dashboard and report generation.

## Final Module Flow

- **Student → Login → Course Tracking → Progress Analysis → Skill Analytics → Recommendation Engine → Dashboard & Reports**

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# MODULE DESIGN

## 1. User & Course Management Module

- **Input:** Student details, login credentials, course information
- **Process:** Validates users and manages student and course records.
- **Output:** Student profile and available course information.
- **Functions:**
  - Student Registration
  - Login Authentication
  - Course Management
  - Learning Activity Management
  - Data Storage



# Module 2: Progress & Skill Analytics



1. **Input:** Quiz scores, completed courses, learning activities
2. **Process:** Analyzes student performance and calculates learning progress.
3. **Output:** Progress status, strengths, and weak areas.
4. **Functions:**
  - Quiz Score Recording
  - Course Completion Tracking
  - Progress Calculation
  - Skill Analysis
  - Strength and Weakness Identification

## 3. Recommendation & Dashboard Module

1. **Input:** Student progress and skill analysis results
2. **Process:** Generates suitable learning recommendations based on the student's performance.
3. **Output:** Personalized learning path, recommendations, and progress reports.
4. **Functions:**
  - Personalized Learning Path Generation
  - Course and Topic Recommendation
  - Weak Area Improvement Suggestions
  - Progress Dashboard
  - Report Generation



Engineer to Excel



# Module Output

## Student Management

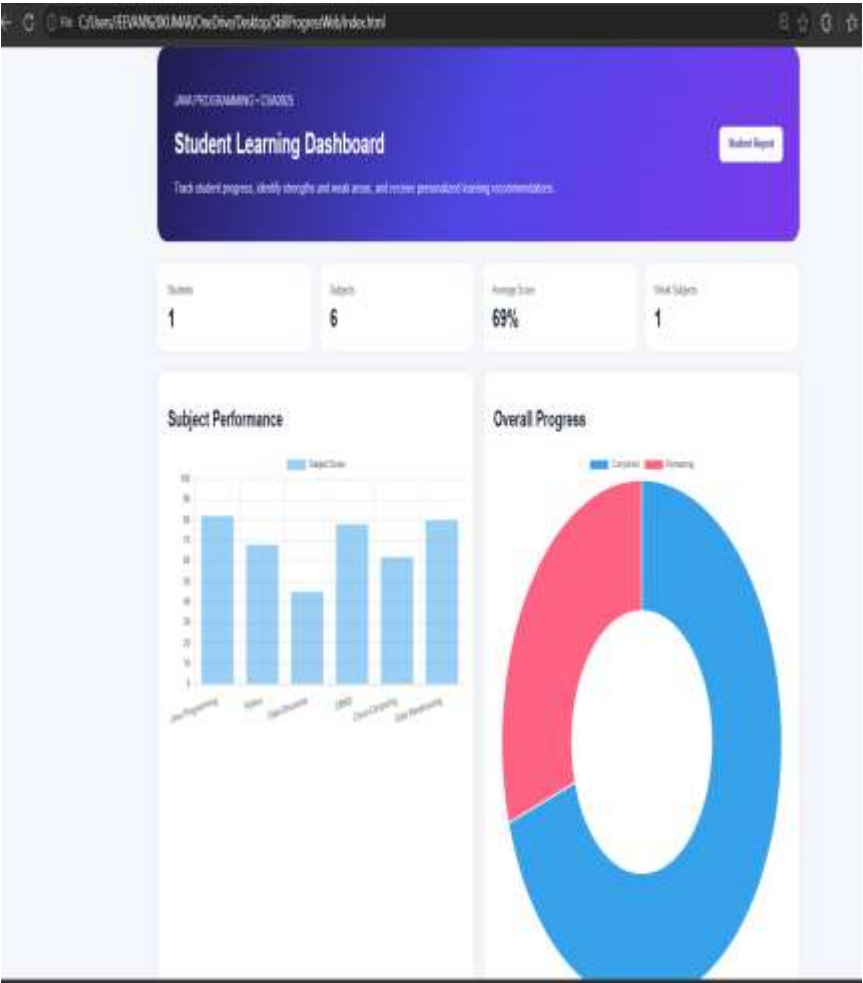
Student registration and student information

### Add Student

Add Student

### Registered Students

| ID | NAME            | EMAIL              | DEPARTMENT | REPORT                      |
|----|-----------------|--------------------|------------|-----------------------------|
| 1  | P. Jeevan Kumar | jeevan@example.com | CSE        | <button>New Report</button> |



## Student Report

Skill progress and personalized learning report

Print Student Report

### Skill Progress Analytics

Personalized Learning Path Recommendation Engine

#### Course Performance

| SUBJECT          | SCORE | PROGRESS | STATUS  |
|------------------|-------|----------|---------|
| Java Programming | 82%   | 85%      | Strong  |
| Python           | 68%   | 70%      | Average |
| Data Structures  | 45%   | 50%      | Weak    |
| DBMS             | 78%   | 75%      | Strong  |
| Cloud Computing  | 62%   | 60%      | Average |
| Data Warehousing | 88%   | 80%      | Strong  |

#### Skill Analysis

| SUBJECT          | SCORE | PERFORMANCE |
|------------------|-------|-------------|
| Java Programming | 82%   | Strong      |
| Python           | 68%   | Average     |
| Data Structures  | 45%   | Weak        |
| DBMS             | 78%   | Strong      |

# UML & Database

## Use Case Diagram

- Student Login
- Student Registration
- View Courses
- Track Progress
- View Recommendations
- Generate Reports

## Class Diagram

- Student
- Course
- Progress
- SkillAnalytics
- Recommendation
- Report

## Sequence Diagram

Login → Validate User → Retrieve Progress → Analyze Skills → Generate Recommendation → Display Result

## ER Diagram

- Student
- Course
- Progress
- Recommendation
- Report



# Technology Stack



## **Programming Language**

Java

## **Database**

MySQL

## **Connectivity**

JDBC

## **Development Tools**

NetBeans / Eclipse

## **Operating System**

Windows 10/11

## **Hardware**

Intel i5/i7 Processor

8 GB RAM

256 GB SSD

# Implementation

## Implementation Progress

- Student registration and login module implemented.
- Course management module developed.
- Progress tracking functionality developed.
- Database structure created using MySQL.
- JDBC connectivity established.
- Recommendation and reporting modules integrated.

## Code Summary

- Java classes handle application logic.
- JDBC handles database communication.
- MySQL stores student and course information.

## Completion:

**Approximately 80–90%**

If your actual completion percentage is different, use your real percentage.



# User Interface



## Login

- Secure student login.
- Registration for new users.
- User validation.

## Dashboard

- Displays student progress.
- Shows skill strengths and weak areas.
- Displays recommended learning paths.

## Reports

- Course completion report.
- Skill analysis report.
- Personalized recommendation report.

## UI Flow:

Login → Dashboard → Progress → Recommendations → Reports

# Testing

## Unit Testing

- Tested login functionality.
- Tested registration functionality.
- Tested progress calculation.
- Tested recommendation module.

## Integration Testing

- Tested Java–JDBC connection.
- Tested JDBC–MySQL communication.
- Tested data flow between modules.

## Expected Outputs

- Correct student login.
- Accurate progress tracking.
- Identification of weak skills.
- Personalized learning recommendations.
- Progress reports.

# Challenges & Future Plan

## Challenges

- Maintaining accurate student learning data.
- Database connectivity issues.
- Integrating multiple Java modules.
- Time limitations during development.

## Solutions

- Added input validation.
- Tested database queries and JDBC connections.
- Integrated modules step-by-step.
- Prioritized core functionalities.

## Next Plan

- Complete final UI improvements.
- Perform complete integration testing.
- Fix remaining errors.
- Improve recommendation accuracy.
- Complete documentation and final deployment.

THANK YOU